

Harry Pan

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EXPERIENCE

Staff Software Engineer

Jun 2025 – Apr 2026

Thoughtful AI

Remote

Designed system architecture to evolve the company's agentic AI platform:

- Architected multi-tenant serverless database cluster enabling onboarding of 8 new customers with secure access from Temporal and Robocloud workloads
- Built provisioning agent that automates tenant onboarding to the database cluster, reducing setup time from 2 hours to 10 minutes
- Designed and implemented analytics lakehouse data pipeline supporting 3 customers
- Built observability pipeline shipping AWS CloudWatch metrics to HyperDX via Kinesis Firehose, establishing a single pane of glass across 5 services
- Enabled data exchange between Thoughtful and sister companies by implementing cross-account Kafka connectivity
- Programming languages: Python, Terraform, SQL

Principal Engineer

Sep 2024 – Jun 2025

Galileo

San Francisco Bay Area

Led system performance optimization initiatives across a platform serving 2M+ daily requests:

- Architected load testing framework and observability stack adopted by 3 engineering teams to establish performance baselines and catch regressions pre-deployment
- Reduced end-to-end API latency by 30% (800ms to 560ms p95) by eliminating redundant database queries through query consolidation and caching
- Decreased task queue latency by 40% by migrating Celery backend from RabbitMQ to Redis, cutting infrastructure costs by \$4K/month
- Increased LLM inference throughput by 80% (120 to 216 req/sec) by implementing dynamic request batching, supporting 2x traffic growth without infrastructure expansion
- Programming languages: Python, Terraform

Principal Engineer

Jul 2023 – Jul 2024

FLYR

San Francisco Bay Area

Led the Cloud Center of Excellence in a cross-functional capacity to establish engineering best practices, enhance efficiency, and drive cost reduction.

- Achieved monthly savings of \$170K through strategic rightsizing and optimization of BigQuery, GCS, CI/CD, and more.
- Delivered cloud unit economics insights to key stakeholders in the C-suite, finance, and accounting departments to inform product pricing and budget planning.
- Enhanced engineering productivity by fostering a culture of ownership and implementing self-service practices.
- Oversaw SaaS vendor contracts, including those with GCP, Confluent Cloud, Datadog, Docker Hub, and others.
- Programming languages: Python, Terraform

Senior Solutions Architect

Dec 2020 – Jun 2023

Amazon Web Services

San Francisco Bay Area

- Scaled startup customer's analytics system to 2x customer capacity while reducing their monthly cloud spend by \$500K
- Selected to AWS Machine Learning Field Committee; guided customer teams in architecting production ML solutions
- Authored [an official AWS ML blog post](#) and contributed to [AWS public code samples](#)
- Built executive-level trust with C-suite and engineering leaders across 10 enterprise accounts
- Programming languages: Python

Software Development Engineer, Alexa AI

Dec 2018 – Dec 2020

Amazon

San Francisco Bay Area

- Co-led development of Alexa AI's Natural Language Understanding MLOps toolchain, used by hundreds of applied scientists, research scientists, and engineers across the org
- Spearheaded the cross-team operational readiness initiative for Amazon's 2019 Prime Day, coordinating 50 individuals across 10 teams
- Partnered with security engineers to review Alexa AI products handling sensitive customer data
- Programming languages: Java

Sr. Software Engineer / Team Lead

Jul 2012 – Nov 2018

Comscore

Portland, OR

Developed distributed systems analyzing billions of TV tuning records daily across a 1M+ line Perl/C++/SQL codebase.

- Developed and maintained in-house distributed systems used to analyze billions of TV tuning records daily.
- Built data verification processes with QA and turned data science prototypes into production products
- Trained data scientists to use OpenGrok to map methodology to code
- Collaborated with infrastructure team to migrate production systems to AWS
- Supported Chinese joint venture Sinotrak in hiring, training, and onboarding data from Chinese cable operators
- Contributed 5K+ lines to OpenGrok and introduced it to the team to navigate and understand the legacy codebase
- Programming languages: Perl, C++, SQL

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript, Scala, Perl, Haskell

Cloud & Infrastructure: AWS, GCP, Kubernetes, Terraform, Docker

Data & ML: LLMs, MLOps, Apache Spark, BigQuery, PostgreSQL, Redis, Celery

Observability & Tools: Datadog, Git, Linux

EDUCATION

Portland State University

Portland, OR

M.S. in Computer Science

Sep 2010 – Jul 2012

Beihang University

Beijing, China

B.S. in Computer Science

Sep 2001 – Jul 2005

CERTIFICATIONS

Certified Kubernetes Administrator

Oct 2023

Coursera – Machine Learning Engineering for Production (MLOps)

Mar 2023

Stanford University – Machine Learning Specialization

Feb 2023

AWS Certified Machine Learning – Specialty

Nov 2022

AWS Certified Solutions Architect – Professional

May 2022

AWS Certified Developer – Associate

Dec 2021

AWS Partner: Cloud Economics Accreditation

Mar 2021